PNAS RESEARCH ARTICLE | PHYSICS SOCIAL SCIENCES OPEN ACCESS

Why more social interactions lead to more polarization in societies

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Edited by David Weitz, Harvard University, Cambridge, MA; received July 7, 2025; accepted October 5, 2025

Over the past two decades, the number of close social connections increased substantially, at least by a factor of two. At the same time, societal opinions have become increasingly polarized in many Western countries. To explore whether these trends could be connected, we employ a simple computational model of society, where people—within their social networks—continuously compare and update their opinions. Here, we show that the model that is known to realistically capture both homophily and social balance exhibits a phase transition phenomenon where, above a critical social connectivity, an explosive transition toward strong polarization must occur. The model allows us to understand the empirical inflation of polarization during the last decades as a function of the observed increased values of social connectivity. In the presence of a small fraction of synchronized influencers, the transition becomes continuous; however, polarization then appears at lower connectivities. We discuss the implications of the presence of a phase transition in social polarization.

polarization | social networks | phase transitions | agent based model | social balance

Significance

In an increasingly interconnected world, rising political polarization threatens social cohesion and democratic stability. Here, we demonstrate that the two trends might be fundamentally connected: Increasing social connectivity, often seen as a unifying force between people, can lead to polarization. Inspired by the physics of collective phenomena, we present a simple model that clarifies the underlying mechanisms. A polarization metric allows us to identify critical density thresholds at which social networks transition from diverse opinions to deeply divided societies. We show that empirically, polarization started to increase exactly with the advent of smartphones and social media. The results might help inspire strategies to mitigate the impact of polarization in digital and real-world communities.

Despite unprecedented levels of social connectivity provided by new forms of communication and social media, many Western societies appear to be more divided than ever. For example, in the United States, growing political polarization increasingly determines how individuals perceive themselves and others (1), and fuels hostility in public discourse (2), strains relationships (3), and influences workplace decisions (4). Political polarization carries well-documented costs such as eroding public trust, undermining democratic processes, and impairing collective problem-solving (5, 6).

Although the existence and extent of polarization were debated over the last two decades ago (7, 8), recently, evidence for an increase and even acceleration of polarization is building up (9–13). For example, the increasing ideological divergence in the US electorate, where Democratic voters shift toward more liberal stances and Republicans toward more conservative ones (14), is visible in survey data of the Pew Research Institute (15); see Fig. 1 *A*–*C*. Clearly, the ideological positions, $\bar{s}_i$, of Democrats and Republicans are seen to diverge between 1999 and 2017. For the way of how ideological positions are quantified, see *Materials and Methods*. Based on the same Pew Research survey (15), the polarization, ψ, shown in Fig. 1*D*, clearly increases sharply from approximately 0.11 before 2010 to 0.19 in 2017. For the definition of the measure of polarization, ψ, see *Materials and Methods* and *SI Appendix*, section 1; for more information on the survey, see *SI Appendix*, section 2.

During the same time period, social connectivity increased considerably. Fig. 1*E* shows the increase of connectivity as reported in a number of surveys conducted in various countries. Here, importantly, social connectivity is defined as the average number of close friendships. The figure reveals a consistent pattern of increasing connectivity starting in 2009 in all the countries (colors), including Germany (17, 18), the Netherlands (18, 19), and Norway (18, 20). Shapes indicate the different surveys. For more details, see *SI Appendix*, section 3. The U.S. surveys (red), all using the identical survey questions, conducted in 2004, 2008, and 2020, are indicated by squares in Fig. 1*E*). All show a clear increase in close friendships (21, 22). The dashed line is a logistic regression to all data points and estimates the average degree of the close friendship network denoted as $\langle k_+ \rangle$. An approximately 100% increase from approximately 2.2 in 2004 to about 4.1 in 2024 is apparent.

This drastic inflation of connectivity in close social ties shortly after 2008 is most likely due to innovations in communication and social media. By 2008, both the iPhone and Android smartphones were launched, and the first version of Facebook was available to the general public [vertical line (I) in panel (*E*)]. The dramatic success of these new media is reflected in the fact that in 2010, Facebook already

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The authors declare no competing interest.

This article is a PNAS Direct Submission.

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This article contains supporting information online at https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2517530122/-/DCSupplemental.

Published October 31, 2025.

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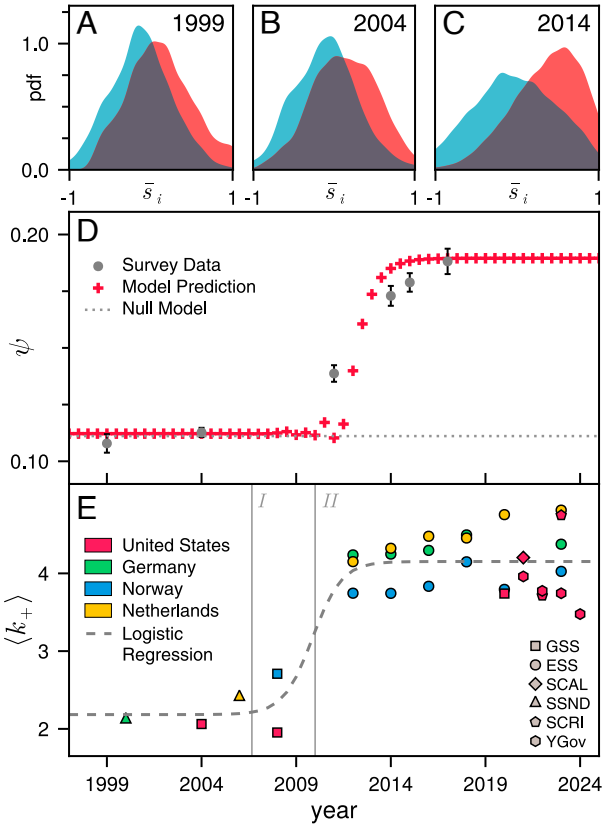

Fig. 1. Increase of polarization and social density over time. (*A–C*) Distribution of ideological opinions in the U.S. electorate. The panels show kernel density estimates of individual ideology, $\bar{s}_i$, for Democrats (blue) and Republicans (red) in 1999, 2004, and 2014 based on Pew Research surveys (15). Clearly, ideological divergence is visible over the years. (*D*) Increase in polarization. Survey-based polarization measure, $\psi = \mathrm{Var}(a_{ij})$, over time (gray circles; error bars denote SD in subsampling of selected survey questions; see *SI Appendix*, section 2). The red line is the model prediction based on parameters $\alpha = 0.5$, $\beta = 2.7$, $\gamma = 0.02$ on a Poissonian small-world with $N = 10^5$ nodes, and a rewiring probability of $\epsilon = 0.175$. The average degree of close friendships, $\langle k_+ \rangle$, is taken from the logistic regression in panel (*E*) (dashed line). The dotted line represents the expected polarization for randomly distributed and noninteracting opinion vectors; see *SI Appendix*, section 1, for details. (*E*) Increase in social connectivity. Estimated average number of close friends, $\langle k_+ \rangle$, by country (colors) and survey (shapes correspond to GSS, ESS, SCAL, SSND, SCRI, YGov; see *SI Appendix*, section 3). The dashed line is a logistic regression through all data points. The transition from low to high connectivity appears shortly after Facebook becomes publicly accessible (vertical line I—2006) and overtakes other websites in US traffic (vertical line II—2010) (16).

had more web traffic than any other US website; see vertical line (II) (16).

Since social connectivity and polarization both increase practically at the same time, it is obvious to ask if both phenomena are related and, if so, how. Here, we address this question by employing a known social opinion dynamics model, where individuals adjust both their opinions and their social ties (23). The reason for this choice is that the model is not only coevolutionary in the sense that the quality of social ties (friendship or enmity) depends on the level of alignment of the individuals' opinions, but it also naturally incorporates two mechanisms that any model of opinion dynamics must include: social balance and homophily.

Homophily, the tendency to befriend like-minded people, plays an important role in opinion dynamics. It can go in both directions: People who are friends tend to align their opinions to become more similar, and people with similar opinions tend to increase the quality of their friendship. Homophily has been shown empirically in many different contexts, including schools (24, 25), workplaces (26), and online communities (27–29), where similar individuals cluster. Once these friendships form, continued interaction typically strengthens their shared beliefs through social influence (30).

The concept of social balance is necessary to understand the possibility that increased connectivity can drive polarization. This might seem counterintuitive, as increased social connectivity would expose individuals to more diverse viewpoints, would create more mutual understanding and tolerance, and thus reduce polarization. However, there is empirical evidence against such simple conclusions. Exposure to opposing political ideas has been shown to often strengthen existing opinions rather than find common ground (31). Any framework trying to explain polarization must be able to capture such negative relations. Importantly, these negative relationships do not form by chance, but emerge in specific patterns. The way they are formed is the essence of the concept of social balance (32), the tendency of three-person groups (triads) to evolve into socially "comfortable" (balanced) configurations. In these balanced configurations, all involved feel a minimum level of cognitive dissonance or social stress. In balanced triads, either all three relations are positive (+++), forming a cohesive friendship group (often with similar opinions), or two relations are negative and one is positive (+ −−), i.e., two friends both dislike a third person. Unbalanced triads create tension: either when three people all dislike each other (− −−) or when one person is friends with two others that dislike each other (++−). Countless empirical studies confirm that balanced triads are significantly overrepresented in social systems (33–38).

The model introduced in ref. 23 combines the reinforcement of similar opinions through homophily and the formation of polarized and fragmented groups through social balance. Both of these components are essential in understanding how increased social connectivity influences polarization. Most earlier models of social opinion dynamics focused on only one of them. For example, voter models (39–42), that only assume positive interactions, usually guarantee consensus under many different conditions. Bounded confidence models (43, 44) that typically only consider homophily predict that low connectivity leads to polarization. More recent models that include more complex social mechanisms, such as antagonistic interactions (45–47) and social balance (23, 48–50). Only recently, the possibility of how increased social connectivity can lead to polarization has been realized (49–52). Among these existing approaches, the model in ref. 23 fulfills the necessary requirements of homophily and social balance in a conceptually simple framework. Moreover, it was found to be compatible with real opinion formation processes in online group experiments (53).

In the model, every individual, i, has an opinion vector, $\mathbf{s}_i$, of length, G. Its binary entries, $s_i^{(k)} = \pm 1$, indicate approval or disapproval of specific issues, k; see Fig. 2*A*. We call the average of $\mathbf{s}_i$ the "ideology" of i and denote it by $\bar{s}_i$. Nodes are connected through a social network that indicates if two individuals know each other, $A_{ij} = 1$, or not, $A_{ij} = 0$. A link between two individuals has a quality, either positive (friendship) or negative (enmity). The network A is fixed, and the link quality (+ or −) can change over time. For A we choose a Poissonian small-world network that has a number of realistic features of tight real-world social networks (54); see *Materials and Methods*. Between any two connected nodes, i and j, we compute their

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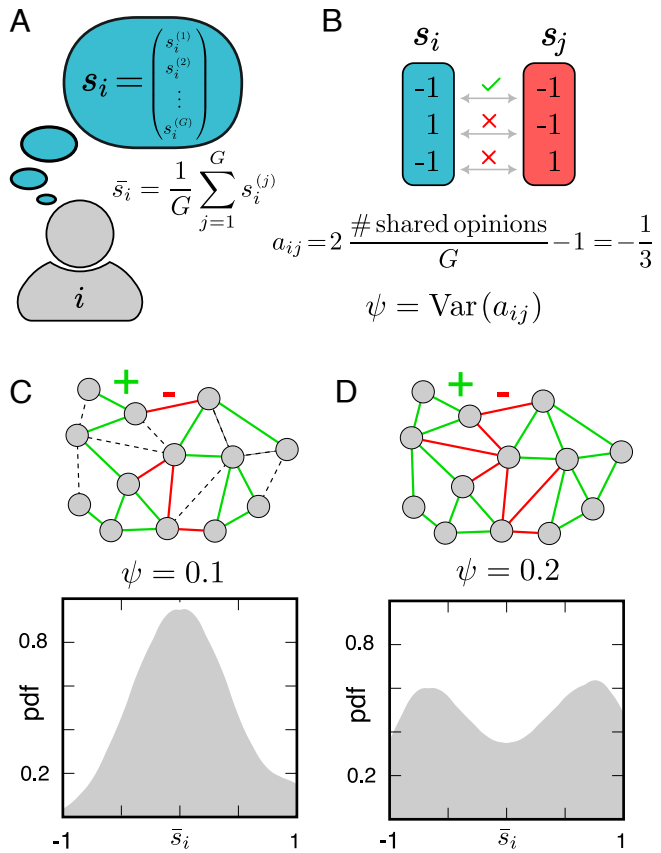

Fig. 2. Schematic view of the polarization measure. (*A*) Opinion vectors and ideology. Each individual, i, holds opinions on G different topics, collected in the vector, $\mathbf{s}_i$ with entries, $s_i^{(j)} = \pm 1$. The ideological position of i is defined as the mean value, $\bar{s}_i$. (*B*) Alignment and polarization. When two people meet, they compare the alignment of their opinion vectors. Example for $G = 3$ with $\mathbf{s}_i = (-1, 1, -1)$, $\mathbf{s}_j = (-1, -1, 1)$. Shared (✓) and opposing opinions (×) yield the normalized alignment of $a_{ij} = -1/3$. Polarization is then calculated as $\psi = \mathrm{Var}(a_{ij})$, where the variance is taken over all pairs (i, j) in the system; see *SI Appendix*, section 1. (*C* and *D*) Example networks at low ($\psi = 0.1$) and medium ($\psi = 0.2$) polarization. Nodes are individuals, green links are friends, red ones are enemies. The histograms show the distributions of $\bar{s}_i$, illustrating more ideological separation at higher ψ.

opinion alignment as the normalized fraction of aligned opinions, $a_{ij} = 2\,(\#\text{shared opinions})/G - 1$. Note that this is the same as the normalized dot product, $a_{ij} = G^{-1}\mathbf{s}_i \cdot \mathbf{s}_j$; see Fig. 2*B*. If $a_{ij} > 0$ the link quality is positive (+), otherwise i and j are enemies (−). Fig. 2 *C* and *D* schematically show how two different signed networks (with positive and negative links) can lead to different polarization, as depicted in the histograms of the ideology distributions.

The cognitive dissonance (or social stress) that any individual feels—given their own opinions and those of their local friends and enemies—is quantified as

$$H^{(i)} = -\alpha \sum_{\{j|j \text{ is friend of } i\}} a_{ij} + (1-\alpha) \sum_{\{j|j \text{ is enemy of } i\}} a_{ij} \quad . \qquad [1]$$

The essence of the model is that every individual tends to minimize their local cognitive dissonance by adapting their opinions such that agreement with friends (first term) and disagreement with enemies (second term) is reached. The parameter α is used to weigh the relevance of the two terms. Typically, one would expect $1 > \alpha \geq 0.5$, which means that it is more important to align with friends than to misalign with enemies.

We implement this model in a Metropolis algorithm; see *Materials and Methods*. In each step, we randomly select one individual, i, and flip one single, randomly chosen, opinion in its opinion vector. We then compute the change in cognitive dissonance, $\Delta H^{(i)}$: If stress decreases, we accept the flip; if stress increases, we accept the update with a certain probability that depends on a parameter, β, that controls the sensitivity of individuals to opinion differences. For details, see *Materials and Methods*. This dynamic ensures that the system can escape local stress minima and converge toward socially balanced configurations. If the flip is accepted and if, as a consequence, the sign of a_{ij} changes, the respective link quality between i and j is updated.

The model is summarized in Fig. 3 *A* and *B*. The histograms show that following the update, ideologies, $\bar{s}_i$, move further apart, reflecting an increase in polarization. Initially, the network in panel (*A*) consists of one balanced and one unbalanced triad. In panel (*B*), the third opinion of node 4 changes from −1 to +1. This now changes the positive link to node 3 to a negative and converts the unbalanced to a balanced triad. Fig. 3*C* reports the cumulative social stress, $H = \sum_i H^{(i)}$, for three values of α and demonstrates that H either decreases from time t to $t+1$, or stays the same, depending on α.

Finally, to be able to study how a small fraction of individuals with fixed, extreme opinions can influence polarization dynamics, we introduce "radicalized influencers" to the model. A fraction γ of individuals is chosen randomly and is assigned fixed extreme

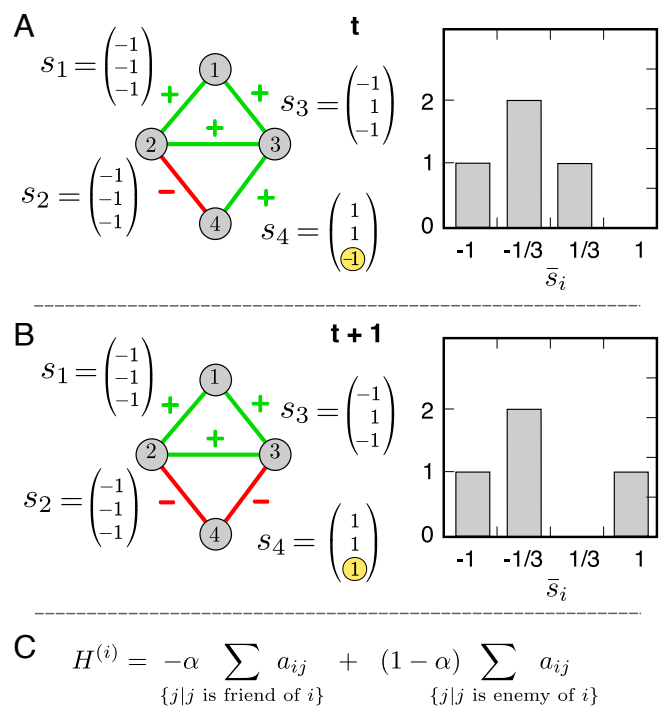

Fig. 3. Opinion and relationship updates. (*A*) State at time t for $G = 3$. Each of the four nodes carries a binary opinion vector, $\mathbf{s}_i$. Edges are green for friendship ($a_{ij} > 0$) and red for disagreement ($a_{ij} < 0$). The histogram shows the four ideological positions $\bar{s}_i = G^{-1}\sum_{j=1}^{3} s_i^{(j)}$ clustered closely. (*B*) State at time $t+1$. Node 4 flips its third opinion. This turns the third and fourth node from friends (green) to enemies (red) and widens the $\bar{s}_i$ distribution. The triad (2, 3, 4) becomes balanced in the process. (*C*) Social stress equation and selected values for cumulative social stress. We calculate the cumulative social stress of the system, $H = \sum_{i=1}^{4} H^{(i)}$. For each value of α, the social stress decreases from time t to $t+1$ while ψ increases.

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opinions; half of them have $s_i^{(k)} = -1$ for all opinions, k, the other half have $s_i^{(k)} = +1$. These nodes do not change their opinions but do update their friendships and enmities based on opinion alignment.

Results

One can immediately infer from the model architecture alone that for high sensitivity, β, and high friendliness, α, the system converges to (+++) triads, forming a large cohesive group with broad agreement and little conflict. For moderate or low α, most triads are in the (+−−) state, fragmenting the population into multiple antagonistic subgroups. Within every subgroup, agreement is strong, but links to most outsiders will be negative. This is the "echo chamber" or "filter bubble" effect, (55, 56). Note that these configurations are still socially balanced through the (+−−) configurations between the bubbles.

We are now ready to test whether increasing social connectivity drives polarization by simulating the model on Poissonian small-world networks (54). By varying the average degree of the underlying networks, we can directly compare the polarization levels from the model predictions (simulations) with the survey-based estimates for every year. See *Materials and Methods* for details of the implementation.

Polarization Transition. In Fig. 4*A*, we present the polarization, ψ, as a function of connectivity (degree) in the network of positive links, $\langle k_+ \rangle$. Green circles show the situation without influencers. A discontinuous transition is visible at a critical degree of $\langle k_+ \rangle \sim 5$, where the system spontaneously transitions to a highly polarized state and continues to polarize more as connectivity increases further. This (phase) transition is also visible in a fundamental social reorganization of group structure. Green circles in Fig. 4*B* show the fraction of nodes in the largest cluster identified by the Leiden algorithm (57).

At the critical degree, many small clusters (each consisting of $\leq$2% of nodes) merge into two opposing groups, each composed of roughly 50% of the population. Below the critical degree, small clusters maintain opinion cohesion while remaining randomly distributed throughout the system, resulting in low overall polarization; see neutral ideologies, $\bar{s}_i$, in the *Left Inset*. Above the critical degree, the two groups show opposing ideologies; see *Right Inset*.

The orange squares in Fig. 4*A* represent the presence of a fraction of 2% influencers, which change the polarization transition into a continuous one. This suggests a disproportionate impact of influencers on polarization dynamics. This qualitative change is again mirrored in the group structure as seen in Fig. 4*B*, orange squares. Community sizes now increase gradually with increasing connectivity. As the transition becomes more gradual, significant polarization occurs earlier at lower connectivities. Also, increasing the sensitivity to social stress, β, leads to an earlier onset of the polarization transition at lower connectivities. The *Inset* of Fig. 4*A* shows the situation for three values of β for the case of influencers present.

Comparison with Empirical Data. The model predicts the empirical polarization trends. As shown in Fig. 1*D*, the empirical polarization in the United States (gray circles) remains comparatively low before 2010, and then rapidly increases from around 0.11 to 0.19 within a few years. The model predicts both the timing and magnitude of the transition remarkably well (red line). In the simulations, only the average degree of the underlying

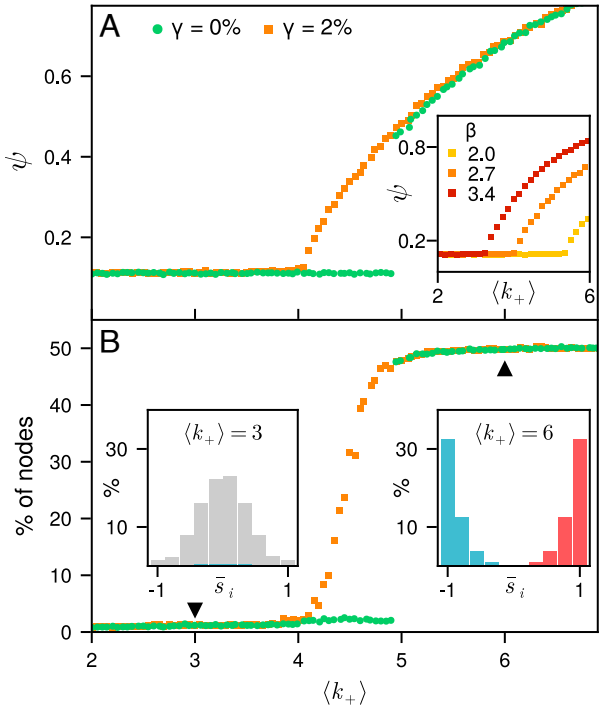

Fig. 4. Polarization transition. (*A*) Polarization, ψ, versus average number of positive ties, $\langle k_+ \rangle$, for simulations without influencers, $\gamma = 0$ (green circles) and with 2% radical influencers, $\gamma = 0.02$ (orange squares). For $\gamma = 0$, ψ remains low until an abrupt jump at the critical connectivity; for $\gamma = 0.02$, ψ increases continuously from lower $\langle k_+ \rangle$. *Inset*: ψ versus $\langle k_+ \rangle$ for $\gamma = 0.02$ at sensitivity values $\beta = 2.00$, $\beta = 2.7$ and $\beta = 3.4$, showing that higher β lead to an earlier transition onset and therefore higher polarization at identical social connectivities, $\langle k_+ \rangle$. (*B*) Fraction of nodes in the largest community [identified by the Leiden algorithm (57)] versus $\langle k_+ \rangle$ for $\gamma = 0$ (green circles) and $\gamma = 0.02$ (orange squares). Above the transition, the largest community contains about ∼50% of the nodes. In all cases, the largest and the second largest groups are approximately identical in size. *Left Inset*: histogram of ideologies, $\bar{s}_i$, of all nodes at $\langle k_+ \rangle = 3$. *Right Inset*: same for $\langle k_+ \rangle = 6$. Each group is displayed in a different color. Simulations parameters: $\alpha = 0.5$, $\beta = 2.7$, Poissonian small-world network with $N = 10^5$ nodes and rewiring probability, $\epsilon = 0.175$.

friendship network is varied according to the empirical rise of social connectivity, as given by the dashed line in Fig. 1*E*. Each prediction (red cross) is calculated using a regression analysis of numerous simulation results, which provides a precise mapping from empirical social connectivity to a polarization value (*SI Appendix*, section 4). All other parameters, $\alpha = 0.5$, $\beta = 2.7$, $\gamma = 0.02$, $N = 10^5$ and $\epsilon = 0.175$ remain fixed. The dotted line in Fig. 1*D* corresponds to a model society with randomly distributed, noninteracting opinions. For details, see *SI Appendix*, section 1.

These parameter values correspond to relatively plausible behavioral assumptions: $\alpha = 0.5$ puts equal weight on aligning with the opinions of friends and avoiding agreement with enemies. We set $\beta = 2.7$. Previous studies on social behavior found that sensitivity parameter β ranges from 1 to 2 for topics like car brands or investments (53). We assume a higher β value for political opinions, reflecting our hypothesis that individuals show more social sensitivity when considering political beliefs. The low fraction of influencers, $\gamma = 0.02$, assumes that fully radicalized individuals are rare. We checked to what extent the model results are robust, qualitatively and quantitatively. We run simulations with slightly varied parameters α, β, γ, and ϵ; see *SI Appendix*, section 5 for the parameter dependence as seen in these robustness checks.

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Hysteresis Effect. The model shows yet another important feature—a hysteresis effect. This means that once social connectivity $\langle k_+ \rangle$ surpasses a critical degree and the network is polarized, the polarization is not immediately reversed by lowering $\langle k_+ \rangle$ to just below the critical threshold. Instead, depolarization requires $\langle k_+ \rangle$ to fall substantially below the threshold at which polarization initially occurred. We demonstrate this hysteresis in *SI Appendix*, section 6 and Fig. S3 *A* and *B*. Notably, in the absence of radical influencers, polarization increases discontinuously with rising $\langle k_+ \rangle$, but decreases continuously with falling $\langle k_+ \rangle$. However, when radical influencers are present, the hysteresis effect is less pronounced: Both polarization and depolarization occur continuously, with only minor differences in their behavior with respect to $\langle k_+ \rangle$ (*SI Appendix*, section 6 and Fig. S3 *C* and *D*).

Discussion

Our results reveal a clear, connectivity-driven mechanism behind rising political polarization: Above a critical social connectivity, an initially cohesive society undergoes a sharp transition into a polarized state. The result is general in the sense that this transition occurs in societies that update opinions and friendship links based on the elementary mechanisms of homophily and social balance. The transition arises solely from the interplay of social balance and homophily. Homophily ensures that similar groups coalesce, while social balance repels opposing groups, resulting in increasingly widening opinion differences.

The main contribution is that the model can be calibrated to survey data that capture the increase in connectivity in real-world social relations over the past two decades. This increased connectivity is related to technical innovations such as the invention of smartphones and the advent of social media, which both appeared around the same year, 2008. Both profoundly changed the way humans communicate immediately after their appearance. The calibrated model correctly predicts both the timing and the approximate extent of the polarization increase in the United States from 2000 to 2020, as quantified in a number of independent surveys , highlighting that rising social connectivity is, by itself, a strong enough mechanism to account for the rise in polarization.

By adding a small number of individuals with extreme opinions (influencers), we observe that the polarization transition becomes continuous. However, large levels of polarization are already reached at connectivities as low as 4.5, which is clearly within the regime of current connectivity levels in Western societies.

The model also contains another potentially relevant message. The polarization transition shows a hysteresis effect, meaning that once polarization occurs as a consequence of increasing connectivity, it can not simply be undone by reversing to previous connectivity levels. This means that once high polarization levels are reached, it becomes increasingly hard to bring them down with policies that would—if such a thing is even thinkable—target to reduce the number of close contacts.

Social network connectivity is not the only driver of polarization in society. One important mechanism has been identified in the increase in political campaign spending. As demonstrated in ref. 58, the 2010 Supreme Court approval of Super PACs that eventually enabled a significant increase in campaign spending did increase polarization in the US legislative chambers. There, the focus was on the emergence of polarization within the US Congress itself. A recent study (59) discusses a similar relation between voter polarization and campaign spending in the context of US House elections. The main finding there is a rapid increase of voter polarization when the campaign spending of both parties exceeds a critical threshold.

The model presented in this paper is not realistic in several aspects, yet it is transparent and contains the most elementary elements in social opinion dynamics. Several parameters can be directly read off from data, such as the degree of social networks, and the value of β (53). Values for α, γ, and ϵ must be estimated: The model dependence on these can be found in the sensitivity analysis summarized in *SI Appendix*, Fig. S3 and section 6.

What is also unrealistic in the present version is to assume that every individual has the same sensitivity to cognitive dissonance, β. Future work making the model more realistic should include a degree of heterogeneity in β and α, which would capture the variation in engagement and tolerance, respectively. Further, it would be desirable to include a more realistic mechanism of social network rewiring, where ties form or dissolve over time and better reflect the fluid nature of social networks. Most social media platforms, for instance, use link recommendation algorithms that inadvertently can further amplify polarization by increasing network clustering (60). However, for all of these improvements to add to the credibility of the model, it would be essential to obtain empirical data on the respective parameters and mechanisms. This we see as the current bottleneck.

Finally, we mention that in the presented model, there is no option to get rid of the polarization transition. What can be done is to shift the critical connectivity (the onset of the transition) in two ways. One is by changing β. Lower β values shift the critical connectivity upward. A β of 2 means a critical polarization onset of a degree well above 5, which is above current connectivities. When β is interpreted as an individual's sensitivity to social stress, a reduction effectively means more "tolerance" toward opposing opinions. The way out of social polarization is either by reducing social contacts (which seems unrealistic) or by increasing tolerance in the society—which might well be achieved by educational means. The other option for shifting the polarization transition is to reduce the disproportionate influence of a small radical minority. This could be achieved through stronger regulatory requirements for content moderation (61) and by disrupting know pathways for radicalization (62). As we have seen, this can increase the critical connectivity threshold and mitigate the onset of polarization. Similarly, building cross-cutting friendships (63)–possibly modeled as a few enforced "bridge" friendships between opposed clusters, or small, targeted network rewiring (64) could lead to a reduction in polarization. Research along these directions is necessary to arrive at the options to proactively counter the increasing levels of polarization.

Materials and Methods

Quantifying Ideological Positions—Opinion Vectors. The ideological position of an individual, i, can be surveyed by collecting a number of her opinions into "opinion vectors," in a collection of G different political opinions of individual i. For every individual, their opinions were collected in an opinion vector, s_i. Every component, $s_i^{(k)}$, reflects a political stance on a particular topic, k, as assessed in ref. 15; see *SI Appendix*, section 2. Opinions on individual stances are binary, in a way that 1 means conservative and −1 liberal. The average over all individual stances and positions, $\bar{s}_i$, defines the "ideology" of the individual. It can be put on an axis from conservative (*Right*) to liberal (*Left*). Fig. 1 *A–C* shows the histogram for the individuals participating in the study; typically, for every different survey, the number of participants varied between 985 and 10,013 individuals; see *SI Appendix*, section 2. The colors denote the self-reported political affiliation of the individuals, with red representing Republicans and those leaning Republican, and blue representing Democrats and those leaning Democrat.

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Measure for Polarization, ψ. We quantify ideological polarization by the alignment variance,

$$\psi = \mathrm{Var}(a_{ij}), \qquad [2]$$

where the alignment between two individuals, i and j is defined as the fraction of shared opinions,

$$a_{ij} = 2\,\frac{\#\{\text{shared opinions between } i \text{ and } j\}}{G} - 1, \qquad [3]$$

where G is the number of all opinions in the opinion vector s_i. Note that this is the same as the normalized dot product of opinion vectors, $a_{ij} = G^{-1}\mathbf{s}_i \cdot \mathbf{s}_i$. This means if two individuals agree on every topic, $a_{ij} = 1$, if they disagree on everything, $a_{ij} = -1$. The alignment variance, ψ, thus ranges from 0 (complete consensus) to 1 (maximal polarization), where the population splits into two equally sized, opposing camps. More information on ψ is provided in *SI Appendix*, section 1.

Model. We build on the social stress framework of ref. 23, where every individual, i, experiences social stress or cognitive dissonance, based on agreement and disagreement with their neighbors. Individual's opinions are represented as G-dimensional vectors $\mathbf{s}_i$, whose components take values in $\{\pm 1\}$. Initially, all $\mathbf{s}_i$ are drawn uniformly at random to arrive at an unbiased starting configuration.

The local social stress of individual i is governed by

$$H^{(i)} = -\frac{\alpha}{G}\sum_{j\in\text{friends}} \mathbf{s}_i\,\mathbf{s}_j + \frac{1-\alpha}{G}\sum_{j\in\text{enemies}} \mathbf{s}_i\,\mathbf{s}_j, \qquad [4]$$

where $\alpha \in [0, 1]$ is the *friendliness parameter*. Higher α favors agreement with friends, lower α encourages opposing enemies.

To highlight how agreement or disagreement shapes $H^{(i)}$ through the pairwise alignment a_{ij}, we transform Eq. **4** into the equivalent form of Eq. **1**. Full details of this transformation appear in *SI Appendix*.

Metropolis Algorithm and Relation Updates. We use the Metropolis algorithm to simulate the opinion dynamics on the network and adapt relationships after an opinion has changed:

1. Node selection: In each iteration, a noninfluencer individual, i, is selected randomly from the network.
2. Proposing new opinion vector: For the selected node, i, a new opinion vector is proposed by flipping one element of the current opinion vector, $s_i^{(k)}$ to its opposite opinion, from $+1$ to -1 or vice versa.
3. Acceptance criterion: The proposed opinion flip is accepted or rejected based on the standard Metropolis acceptance criterion $P_{\text{accept}} = \min\left\{1, \exp(-\beta\Delta H_i)\right\}$, where $\Delta H_i = H_i^{\text{after flip}} - H_i^{\text{before flip}}$.
4. Updating relations: If the new opinion vector is accepted, the relations between the selected individual, i, and their neighbors, j, are updated, according to $\mathrm{sign}(\mathbf{s}_i \cdot \mathbf{s}_j)$. A positive sign indicates friendship, a negative sign means enmity.

The parameter β governs an individual's responsiveness to social stress when flipping opinions. $\beta = 0$ means that stress is ignored, and changes are random; as β increases, individuals increasingly favor changes that reduce stress. Following ref. 65, we interpret β as the sensitivity to opinion differences.

Poissonian Small-World Network. In all simulations, we use a Poissonian small-world network to model social connectivity. Networks of this type are simple and realistic for modeling tight relationships and have been used in epidemic models (54).

To construct a Poissonian small-world network, N nodes are placed on a ring, initially without edges. Every node is assigned a target degree based on a Poisson distribution with mean λ, corresponding to the desired average degree of the network $\langle k\rangle$. Edges are then added, with nodes preferentially connecting to their nearest neighbors, until the assigned degree of every node is matched.

Following the initial edge assignment, we rewire the network according to the small-world algorithm (66), with a rewiring probability, ϵ. For small ϵ, this introduces shortcuts, while preserving the network's overall structure. We exclude networks that contain isolated nodes. Once the simulation of opinion dynamics reaches a steady state, we measure the average degree $\langle k_+\rangle$ of the network consisting only of friendly connections. Throughout the paper, we use $\epsilon = 0.175$

Data, Materials, and Software Availability. Raw survey and connectivity data are available from the sources detailed in *SI Appendix*. Cleaned, filtered datasets, and all code necessary to reproduce the analyses and simulations have been deposited in GitHub (https://github.com/complexity-sciencehub/ConnectivityPolarizationTransition). Previously published data were used for this work (15).

ACKNOWLEDGMENTS. We acknowledge support from the Austrian Science Fund (FWF) under Grants No. 10.55776/P34994 and EFP5 ReMASS, funding from the Austrian Federal Ministry for Climate Action, Environment, Energy, Mobility, Innovation, and Technology under GZ 2023-0.841.266, through the Postdoc Program for Complexity Science and Data Competence.

1. J. T. Jost, D. S. Baldassarri, J. N. Druckman, Cognitive-motivational mechanisms of political polarization in social-communicative contexts. *Nat. Rev. Psychol.* **1**, 560–576 (2022).
2. J. Benson, Democracy and the epistemic problems of political polarization. *Am. Polit. Sci. Rev.* **118**, 1719–1732 (2024).
3. W. J. Chopik, M. Motyl, Ideological fit enhances interpersonal orientations. *Soc. Psychol. Pers. Sci.* **7**, 759–768 (2016).
4. K. Gift, T. Gift, Does politics influence hiring? Evidence from a randomized experiment *Polit. Behav.* **37**, 653–675 (2015).
5. E. J. Finkel *et al.*, Political sectarianism in America. *Science* **370**, 533–536 (2020).
6. M. H. Graham, M. W. Svolik, Democracy in America? Partisanship, polarization, and the robustness of support for democracy in the United States *Am. Polit. Sci. Rev.* **114**, 392–409 (2020).
7. M. P. Fiorina, *Culture War? the Myth of a Polarized America* (Pearson Longman, New York, 2005).
8. A. I. Abramowitz, K. L. Saunders, Is polarization a Myth? *J. Polit.* **70**, 542–555 (2008).
9. S. Iyengar, G. Sood, Y. Lelkes, Affect, not ideology. *Public Opin. Q.* **76**, 405–431 (2012).
10. S. Iyengar, S. J. Westwood, Fear and loathing across party lines: New evidence on group polarization. *Am. J. Polit. Sci.* **59**, 690–707 (2015).
11. A. I. Abramowitz, S. Webster, The rise of negative partisanship and the nationalization of U.S. elections in the 21st century. *Elect. Stud.* **41**, 12–22 (2016).
12. S. Iyengar, Y. Lelkes, M. Levendusky, N. Malhotra, S. J. Westwood, The origins and consequences of affective polarization in the United States. *Annu. Rev. Polit. Sci.* **22**, 129–146 (2019).
13. J. Phillips, Affective polarization: Over time, through the generations, and during the lifespan. *Polit. Behav.* **44**, 1483–1508 (2022).
14. S. W. Webster, A. I. Abramowitz, The ideological foundations of affective polarization in the U.S. *Electorate. Am. Polit. Res.* **45**, 621–647 (2017).
15. Pew Research Center, Datasets Archive (2025). https://www.pewresearch.org/datasets/. Accessed 1 May 2025.
16. R. E. Wilson, S. D. Gosling, L. T. Graham, A review of Facebook research in the social sciences. *Perspect. Psychol. Sci.* **7**, 203–220 (2012).
17. T. Wöhler, T. Hinz, Core discussion in the USA and Germany. *Kolner Z. Soziol. Sozialpsychol.* **59**, 91–112+520 (2007).
18. European Social Survey, European social survey: Rounds 6–11 (2012–2023). https://ess.sikt.no/en/. Accessed 15 April 2025.
19. G. Mollenhorst, B. Volker, H. Flap, Changes in personal relationships: How social contexts affect the emergence and discontinuation of relationships. *Soc. Netw.* **37**, 65–80 (2014).
20. K. N. Hampton, R. Ling, Explaining communication displacement and large-scale social change in core networks: A cross-national comparison of why bigger is not better and less can mean more. *Inf. Commun. Soc.* **16**, 561–589 (2013).
21. M. McPherson, L. Smith-Lovin, M. E. Brashears, Social isolation in America: Changes in core discussion networks over two decades. *Am. Sociol. Rev.* **71**, 353–375 (2006).
22. N. Pennington, J. A. Hall, A. J. Holmstrom, The American friendship project: A report on the status and health of friendship in America. *PLoS One* **19**, e0305834 (2024).
23. T. M. Pham, J. Korbel, R. Hanel, S. Thurner, Empirical social triad statistics can be explained with dyadic homophylic interactions. *Proc. Natl. Acad. Sci. U.S.A.* **119**, e2121103119 (2022).
24. W. Shrum, N. H. Cheek, S. M. Hunter, Friendship in school: Gender and racial homophily. *Sociol. Educ.* **61**, 227 (1988).
25. I. Smirnov, S. Thurner, Formation of homophily in academic performance: Students change their friends rather than performance. *PLoS One* **12**, e0189564 (2017).
26. A. M. Kleinbaum, T. E. Stuart, M. L. Tushman, Discretion within constraint: Homophily and structure in a formal organization. *Organ. Sci.* **24**, 1316–1336 (2013).
27. G. Kossinets, D. J. Watts, Origins of homophily in an evolving social network. *Am. J. Sociol.* **115**, 405–450 (2009).
28. L. M. Aiello *et al.*, Friendship prediction and homophily in social media. *ACM Trans. Web* **6**, 1–33 (2012).
29. K. Z. Khanam, G. Srivastava, V. Mago, The homophily principle in social network analysis: A survey. *Multimed. Tools Appl.* **82**, 8811–8854 (2023).

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30. R. B. Cialdini, N. J. Goldstein, Social influence: Compliance and conformity. *Annu. Rev. Psychol.* **55**, 591–621 (2004).
31. C. A. Bail *et al.*, Exposure to opposing views on social media can increase political polarization. *Proc. Natl. Acad. Sci. U.S.A.* **115**, 9216–9221 (2018).
32. F. Heider, Attitudes and cognitive organization. *J. Psychol.* **21**, 107–112 (1946).
33. H. Taylor, *Balance in Small Groups* (Van Nostrand Reinhold Company, 1979).
34. J. Leskovec, D. Huttenlocher, J. Kleinberg, "Signed networks in social media" in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, E. Mynatt, S. E. Hudson, G. Fitzpatrick, W. K. Edwards, T. Rodden, Eds. (ACM, Atlanta, Georgia, USA, 2010), pp. 1361–1370.
35. M. Szell, S. Thurner, Measuring social dynamics in a massive multiplayer online game. *Soc. Netw.* **32**, 313–329 (2010).
36. G. Facchetti, G. Iacono, C. Altafini, Computing global structural balance in large-scale signed social networks. *Proc. Natl. Acad. Sci. U.S.A.* **108**, 20953–20958 (2011).
37. X. Zheng, D. Zeng, F. Y. Wang, Social balance in signed networks. *Inf. Syst. Front.* **17**, 1077–1095 (2011).
38. A. Kirkley, G. T. Cantwell, M. E. J. Newman, Balance in signed networks. *Phys. Rev. E* **99**, 012320 (2019).
39. P. Clifford, A. Sudbury, A model for spatial conflict. *Biometrika* **60**, 581–588 (1973).
40. R. A. Holley, T. M. Liggett, Ergodic theorems for weakly interacting infinite systems and the voter model. *Ann. Probab.* **3**, 643–663 (1975).
41. F. Vazquez, V. M. Eguiluz, Generic absorbing transition in coevolution dynamics. *Phys. Rev. Lett.* **100**, 108702 (2008).
42. S. Redner, *A Guide to First-Passage Processes* (Cambridge University Press, 2001).
43. G. Deffuant, D. Neau, F. Amblard, G. Weisbuch, Mixing beliefs among interacting agents. *Adv. Complex Syst.* **03**, 87–98 (2000).
44. H. Rainer, U. Krause, Opinion dynamics and bounded confidence: Models, analysis and simulation. *J. Artif. Soc. Soc. Simul.* **5**, 121900 (2002).
45. S. Biswas, A. Chatterjee, P. Sen, Disorder induced phase transition in kinetic models of opinion dynamics. *Phys. A Stat. Mech. Appl.* **391**, 3257–3265 (2012).
46. G. Shi, A. Proutiere, M. Johansson, J. S. Baras, K. H. Johansson, The evolution of beliefs over signed social networks. *Oper. Res.* **64**, 585–604 (2016).
47. D. Bhat, S. Redner, Polarization and consensus by opposing external sources. *J. Stat. Mech.* **20**, 013402 (2020).
48. C. Altafini, Dynamics of opinion forming in structurally balanced social networks. *PLoS One* **7**, e38135 (2012).
49. T. Minh Pham, I. Kondor, R. Hanel, S. Thurner, The effect of social balance on social fragmentation. *J. R. Soc. Interface* **17**, 20200752 (2020).
50. T. M. Pham, A. C. Alexander, J. Korbel, R. Hanel, S. Thurner, Balance and fragmentation in societies with homophily and social balance. *Sci. Rep.* **11**, 17188 (2021).
51. O. Sikder, R. E. Smith, P. Vivo, G. Livan, A minimalistic model of bias, polarization and misinformation in social networks. *Sci. Rep.* **10**, 5493 (2020).
52. T. Pham, S. Redner, L. Waldorp, J. Armas, H. L. J. van der Maas, Polarisation in increasingly connected societies. arXiv [Preprint] (2025). https://arxiv.org/abs/2503.24098
53. M. Galesic, H. Olsson, T. M. Pham, J. Sorger, S. Thurner, Experimental evidence confirms that triadic social balance can be achieved through dyadic interactions. *NPJ Complex.* **2**, 1 (2025).
54. S. Thurner, P. Klimek, R. Hanel, A network-based explanation of why most COVID-19 infection curves are linear. *Proc. Natl. Acad. Sci. U.S.A.* **117**, 22684–22689 (2020).
55. M. Cinelli, G. De Francisci Morales, A. Galeazzi, W. Quattrociocchi, M. Starnini, The echo chamber effect on social media. *Proc. Natl. Acad. Sci. U.S.A.* **118**, e2023301118 (2021).
56. S. Flaxman, S. Goel, J. M. Rao, Filter bubbles, echo chambers, and online news consumption. *Public Opin. Q.* **80**, 298–320 (2016).
57. V. A. Traag, L. Waltman, N. J. Van Eck, From Louvain to Leiden: Guaranteeing well-connected communities. *Sci. Rep.* **9**, 5233 (2019).
58. X. Lu, J. Gao, B. K. Szymanski, The evolution of polarization in the legislative branch of government. *J. R. Soc. Interface* **16**, 20190010 (2019).
59. J. Korbel, R. Dahdoul, S. Thurner, Campaign-spending driven polarization transition in a double-random field model of elections. arXiv [Preprint] (2025). https://arxiv.org/abs/2510.00612
60. F. P. Santos, Y. Lelkes, S. A. Levin, Link recommendation algorithms and dynamics of polarization in online social networks. *Proc. Natl. Acad. Sci. U.S.A.* **118**, e2102141118 (2021).
61. A. Trujillo, T. Fagni, S. Cresci, The DSA transparency database: Auditing self-reported moderation actions by social media. *Proc. ACM Hum. Comput. Interact.* **9**, 1–28 (2025).
62. M. H. Ribeiro, R. Ottoni, R. West, V. A. F. Almeida, W. Meira, "Auditing radicalization pathways on YouTube" in *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, M. Hildebrandt *et al.*, Eds. (ACM, Barcelona Spain), pp. 131–141 (2020).
63. S. Balietti, L. Getoor, D. G. Goldstein, D. J. Watts, Reducing opinion polarization: Effects of exposure to similar people with differing political views. *Proc. Natl. Acad. Sci. U.S.A.* **118**, e2112552118 (2021).
64. R. Axelrod, J. J. Daymude, S. Forrest, Preventing extreme polarization of political attitudes. *Proc. Natl. Acad. Sci. U.S.A.* **118**, e2102139118 (2021).
65. M. Galesic, H. Olsson, J. Dalege, T. Van Der Does, D. L. Stein, Integrating social and cognitive aspects of belief dynamics: towards a unifying framework. *J. R. Soc. Interface* **18**, 20200857 (2021).
66. D. J. Watts, S. H. Strogatz, Collective dynamics of 'small-world' networks. *Nature* **393**, 440–442 (1998).

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